

Question 1

T1

Liam went for a ride on a Ferris wheel with a radius of 5 metres. Determine the distance he travelled if the Ferris wheel stopped after rotating 1260° .

Solution

$$\theta = 1260^\circ \cdot \frac{\pi}{180^\circ} \quad 1 \text{ mark for conversion}$$

$$\theta = 7\pi$$

or

$$\theta = 21.991148\dots$$

$$s = \theta r$$

$$s = 7\pi(5) \quad 1 \text{ mark for substitution}$$

$$s = 35\pi \text{ metres}$$

or

$$s = 109.956 \text{ metres}$$

2 marks

Determine and simplify the term that contains x^{12} in the binomial expansion of $(2x^4 - 3)^7$.

Solution

Method 1

$$(x^4)^7, (x^4)^6, (x^4)^5, \dots$$

$$x^{28}, x^{24}, x^{20}, \dots$$

1 mark for determining a pattern

$$t_5 = {}_7C_4 (2x^4)^3 (-3)^4$$

$$= 22\,680x^{12}$$

2 marks (1 mark for ${}_7C_4$; ½ mark for each consistent factor)

1 mark for simplification (½ mark for coefficient, ½ mark for exponent)

4 marks

Method 2

$$x^{12} = (x^4)^{7-k} (x^0)^k$$

½ mark for substitution

$$12 = 28 - 4k$$

$$-16 = -4k$$

$$4 = k$$

½ mark for solving for k

$$t_5 = {}_7C_4 (2x^4)^3 (-3)^4$$

$$= 22\,680x^{12}$$

2 marks (1 mark for ${}_7C_4$; ½ mark for each consistent factor)

1 mark for simplification (½ mark for coefficient, ½ mark for exponent)

4 marks

Question 3

R10

The normal healing of wounds can be modeled by an exponential function. The area of the wound decreases according to the formula,

$$A = A_0 e^{-0.35t}$$

where: A is the area of the wound in cm^2 , after t days,
 A_0 is the initial area of the wound in cm^2 , and
 t is the time, in days, after healing begins.

If a wound has an initial area of 25 cm^2 , determine:

- a) the area of the wound after 3 days.
- b) the amount of time it will take for the area of the wound to decrease to 4 cm^2 .

Solution

a) $A = 25e^{-0.35(3)}$

$$A = 8.748 \text{ cm}^2$$

1 mark

b) $4 = 25e^{-0.35t}$

$$0.16 = e^{-0.35t}$$

$$\ln 0.16 = \ln e^{-0.35t}$$

1 mark for applying logarithms

$$\ln 0.16 = -0.35t \ln e$$

½ mark for power law

$$\frac{\ln 0.16}{-0.35} = t$$

$$t = 5.235947...$$

$$t = 5.236 \text{ days}$$

½ mark for the value of t

2 marks

Question 4

P3

During a hockey practice, 4 players are wearing a green jersey and 5 players are wearing a red jersey. Determine the number of ways three players can be selected if at least two of the players must be wearing a red jersey.

Solution

Case 1: ${}_5C_2 \cdot {}_4C_1 = 40$

1 mark for Case 1

Case 2: ${}_5C_3 \cdot {}_4C_0 = 10$

1 mark for Case 2

$40 + 10 = 50$ ways

1 mark for addition of cases

3 marks

Note:

${}_4C_0$ does not need to be shown.

Question 5

T5

Solve, algebraically, over the interval $[0, 2\pi]$.

$$2\sin^2 x + 5\cos x + 1 = 0$$

Solution

$$2(1 - \cos^2 x) + 5\cos x + 1 = 0$$

1 mark for substitution of an appropriate identity

$$2 - 2\cos^2 x + 5\cos x + 1 = 0$$

$$-2\cos^2 x + 5\cos x + 3 = 0$$

$$2\cos^2 x - 5\cos x - 3 = 0$$

$$(2\cos x + 1)(\cos x - 3) = 0$$

$$\cos x = -\frac{1}{2} \quad \cos x = 3$$

1 mark for solving for $\cos x$

$$x = \frac{2\pi}{3}, \frac{4\pi}{3} \quad \text{no solution}$$

2 marks for solving for x ($\frac{1}{2}$ mark for each value; 1 mark for indicating no solution)

4 marks

Question 6

R3

The function, $y = f(x)$, has a domain of $[-8, 4]$. The domain of the graph of $y = f(bx)$ is $[-4, 2]$.

State the value of b .

Solution

$$b = 2$$

1 mark

Question 7**R14**

Describe the difference between the graph of $f(x) = \frac{(x+3)(x-4)}{(x+3)}$ and the graph of $g(x) = x - 4$.

Solution

The graph of $f(x)$ has a point of discontinuity at $x = -3$, where as the graph of $g(x)$ does not.

1 mark

Question 8

T2

Determine if the point $\left(-\frac{3}{5}, \frac{2}{\sqrt{3}}\right)$ is on the unit circle. Justify your answer.

Solution

Method 1

$$x^2 + y^2 = 1$$

$$\begin{aligned}\text{Left-Hand Side} &= \left(-\frac{3}{5}\right)^2 + \left(\frac{2}{\sqrt{3}}\right)^2 \\ &= \frac{9}{25} + \frac{4}{3} \\ &= \frac{27}{75} + \frac{100}{75}\end{aligned}$$

$$\frac{127}{75} \neq 1$$

Therefore the point $\left(-\frac{3}{5}, \frac{2}{\sqrt{3}}\right)$ is not on the unit circle.

1 mark for justification

1 mark

Method 2

Since $\frac{2}{\sqrt{3}} > 1$, the point is not on the unit circle.

1 mark for justification

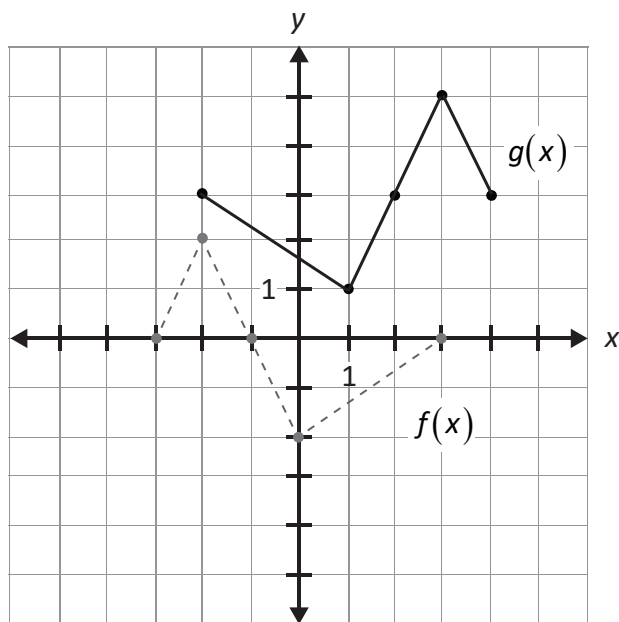
1 mark

Question 9

R2, R5

Given the graph of $f(x)$, sketch the graph of $g(x) = f(-(x-1)) + 3$.

Solution



1 mark for horizontal reflection
1 mark for horizontal translation
1 mark for vertical translation

3 marks

Question 10

R11

Bonnie was asked to determine the factors of $p(x) = -3x^4 + 2x^2 - 4x + 5$.

Describe the error she made in her set up of synthetic division.

$$\begin{array}{r|rrrrr} 1 & -3 & 2 & -4 & 5 & \\ & & & & & \end{array}$$

Solution

Bonnie did not write the coefficient of 0 for the missing term, x^3 .

1 mark

Question 11

R1

Given the functions, $f(x) = \frac{1}{x-1}$ and $g(x) = x - 3$,

a) state the equation of $f(g(x))$.

b) state the domain of $f(g(x))$.

Solution

a) $f(g(x)) = \frac{1}{x-4}$

1 mark

b) Domain: $\{x \mid x \neq 4, x \in \mathbb{R}\}$

or

Domain: $(-\infty, 4) \cup (4, \infty)$

Question 12

T6

Prove the identity for all permissible values of x .

$$\tan 2x = \frac{2}{\cot x - \tan x}$$

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\tan 2x$	$\frac{2}{\frac{\cos x}{\sin x} - \frac{\sin x}{\cos x}}$ $\frac{2}{\frac{\cos^2 x - \sin^2 x}{\sin x \cos x}}$ $\frac{2 \sin x \cos x}{\cos^2 x - \sin^2 x}$ $\frac{\sin 2x}{\cos 2x}$ $\tan 2x$

1 mark for correct substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 12

T6

Method 2

Left-Hand Side	Right-Hand Side
$\frac{2 \tan x}{1 - \tan^2 x}$	$\frac{2}{\frac{\cos x}{\sin x} - \frac{\sin x}{\cos x}}$
$\frac{\frac{2 \sin x}{\cos x}}{1 - \frac{\sin^2 x}{\cos^2 x}}$	$\frac{2}{\frac{\cos^2 x - \sin^2 x}{\sin x \cos x}}$
$\frac{\frac{2 \sin x}{\cos x}}{\frac{\cos^2 x - \sin^2 x}{\cos^2 x}}$	$\frac{2 \sin x \cos x}{\cos^2 x - \sin^2 x}$
$\frac{2 \sin x \cos x}{\cos^2 x - \sin^2 x}$	

1 mark for correct substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 12

T6

Method 3

Left-Hand Side	Right-Hand Side
$\frac{2 \tan x}{1 - \tan^2 x}$	$\frac{2}{\frac{1}{\tan x} - \tan x}$ $\frac{2}{\frac{1 - \tan^2 x}{\tan x}}$ $\frac{2 \tan x}{1 - \tan^2 x}$

1 mark for correct substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 13

R1

Given $f(x) = -x^2 + 2$ and $g(x) = x^2 - 1$,

a) determine the equation $h(x) = g(x) - f(x)$.

b) state the range of $h(x)$.

Solution

a) $h(x) = g(x) - f(x)$

1 mark for subtraction of $g(x) - f(x)$

$$h(x) = (x^2 - 1) - (-x^2 + 2)$$

$$h(x) = x^2 - 1 + x^2 - 2$$

$$h(x) = \underline{2x^2 - 3}$$

1 mark

b) Range: $\underline{\{y | y \geq -3, y \in \mathbb{R}\}}$

1 mark for range consistent with $h(x)$

or

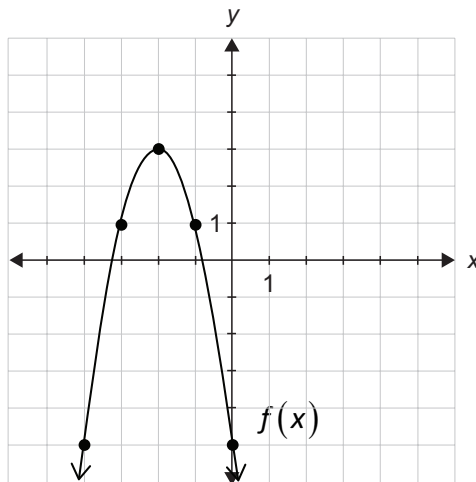
1 mark

Range: $\underline{[-3, \infty)}$

Question 14

R5, R6

Given the graph of $f(x)$,

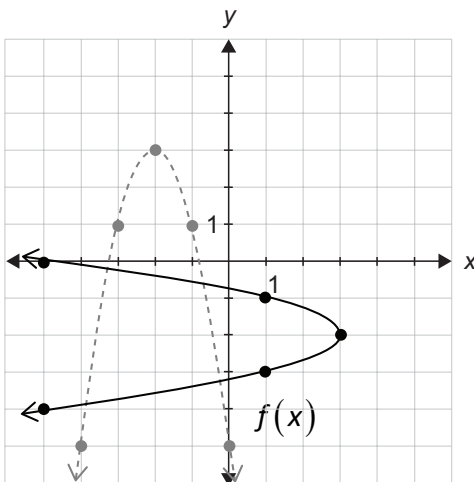


a) sketch the graph after a reflection over the line $y = x$.

b) state a restriction on the domain of $f(x)$ in order for its inverse to be a function.

Solution

a)



1 mark

b) $\{x \mid x \geq -2, x \in \mathbb{R}\}$

or

$\{x \mid x \leq -2, x \in \mathbb{R}\}$

1 mark

Note:

Other answers are possible for b).

Question 15

R4, R5

Describe a transformation of a function that would not affect its domain.

Solution

vertical stretch by a factor of a

or

vertical reflection

or

vertical translation k units up/down

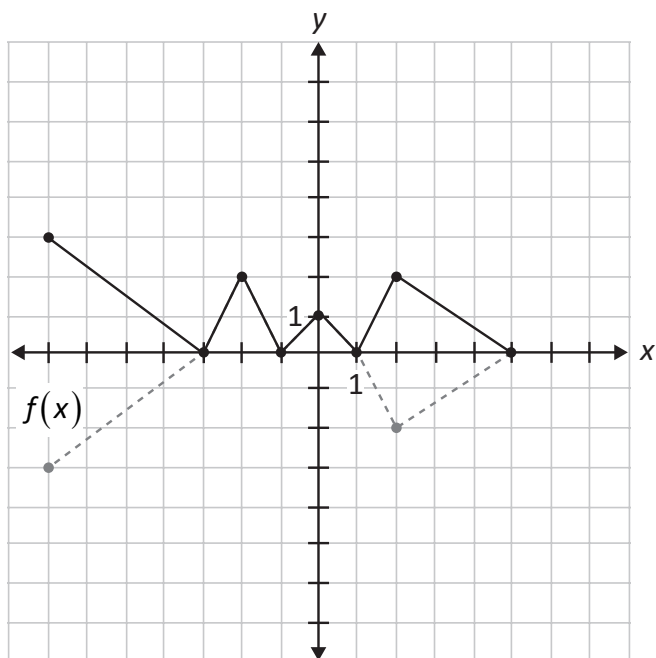
1 mark

Question 16

R1

Given the graph of $f(x)$, sketch the graph of the function $y = |f(x)|$.

Solution



1 mark

Question 17

R11

Determine, algebraically, all of the zeros of the polynomial function, $p(x) = 3x^3 + x^2 - 20x + 12$.

Solution

$$p(x) = 3x^3 + x^2 - 20x + 12$$

$$p(2) = 3(2)^3 + (2)^2 - 20(2) + 12$$

$$p(2) = 0$$

1 mark for identifying one zero of $p(x)$

$\therefore (x - 2)$ is a factor

2	3	1	-20	12
		6	14	-12
	3	7	-6	0

1 mark for synthetic division (or equivalent strategy)

$$p(x) = (x - 2)(3x^2 + 7x - 6)$$

$$0 = (x - 2)(3x - 2)(x + 3)$$

$\frac{1}{2}$ mark for consistent factors

$$x = 2, \quad x = \frac{2}{3}, \quad x = -3$$

$\frac{1}{2}$ mark for consistent zeros

3 marks

Question 18**T2**

Given $\theta = 240^\circ$, identify the coordinates of the point, $P(\theta)$, on the unit circle.

a. $\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

b. $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$

c. $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

d. $\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

Question 19**R1**

Given the functions, $f(x) = -3x + 5$ and $g(x) = x^2 + x - 1$, identify the value of $g(f(2))$.

a. -10

b. -5

c. -3

d. -1

Question 20**R3**

Given $f(x) = \sqrt{x} + 5$, identify the equation of the transformed graph that has the same y -intercept.

a. $y = f(x-3)$

b. $y = -3f(x)$

c. $y = f(-3x)$

d. $y = f(x) - 3$

Question 21**R12**

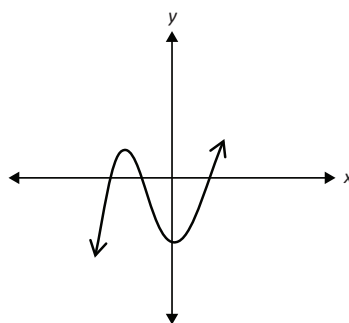
Given the graph of $p(x) = ax^3 + bx^2 + cx + d$, identify the statement that is true.

a. $a > 0, d < 0$

b. $a > 0, d > 0$

c. $a < 0, d < 0$

d. $a < 0, d > 0$

**Question 22****T5**

Identify the general solution of the equation, $\csc\theta = -1$.

a. $\theta = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$

b. $\theta = \frac{3\pi}{2} + 2k\pi, k \in \mathbb{Z}$

c. $\theta = \frac{3\pi}{2} + k\pi, k \in \mathbb{Z}$

d. $\theta = \pi + 2k\pi, k \in \mathbb{Z}$

Question 23**R13**

Identify the equation of a radical function with a domain of $[-6, \infty)$ and a range of $(-\infty, 3]$.

a. $y = -\sqrt{x+6} - 3$

b. $y = -\sqrt{x+6} + 3$

c. $y = \sqrt{-(x+3)} - 6$

d. $y = \sqrt{x-6} + 3$

Question 24**P3**

Identify the value of n in the equation, ${}_nC_3 = {}_nC_7$.

a. 3

b. 4

c. 7

d. 10

Question 25**R11**

Identify the remainder when $p(x) = x^5 - 1$ is divided by $(x+1)$.

a. -2

b. -1

c. 0

d. 4

Question 26

R13

Match each function with its corresponding graph.

Solution

Write the corresponding letter in this column.

$$y = \sqrt{x} - 3$$

 B

$$y = \sqrt{-x} + 3$$

 D

$$y = -\sqrt{x+3}$$

 C

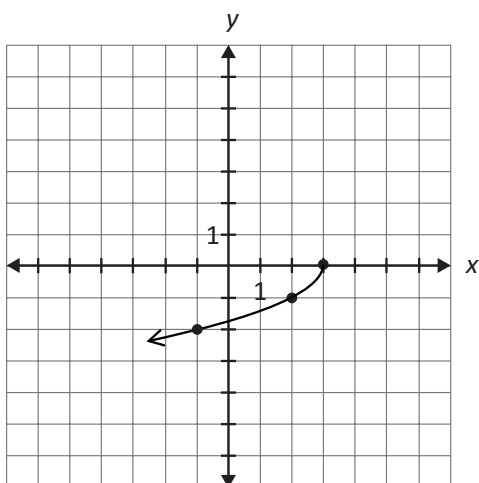
$$y = -\sqrt{-(x-3)}$$

 A

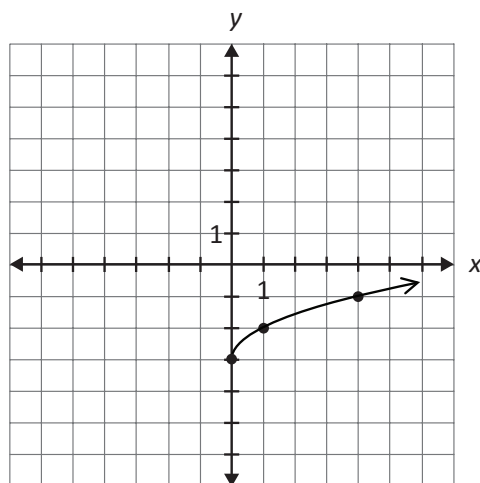
½ mark for each correct answer

2 marks

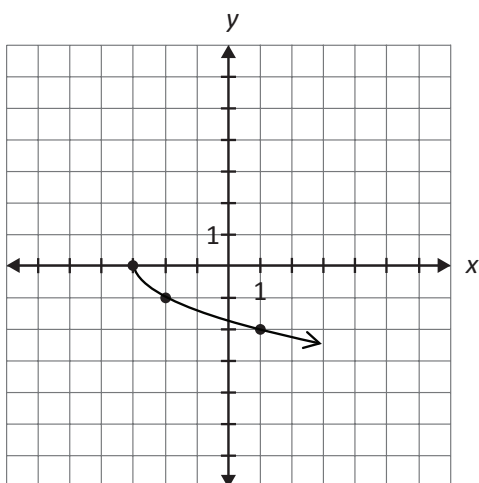
A)



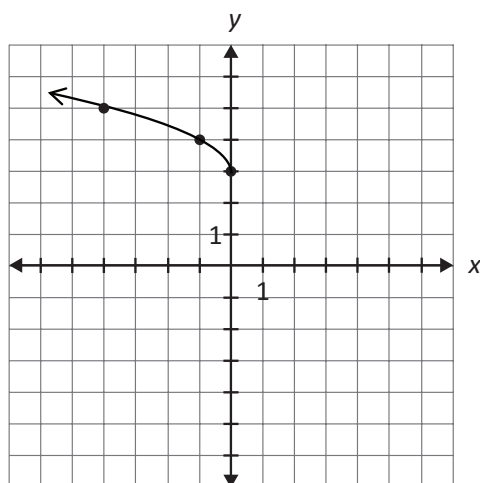
B)



C)



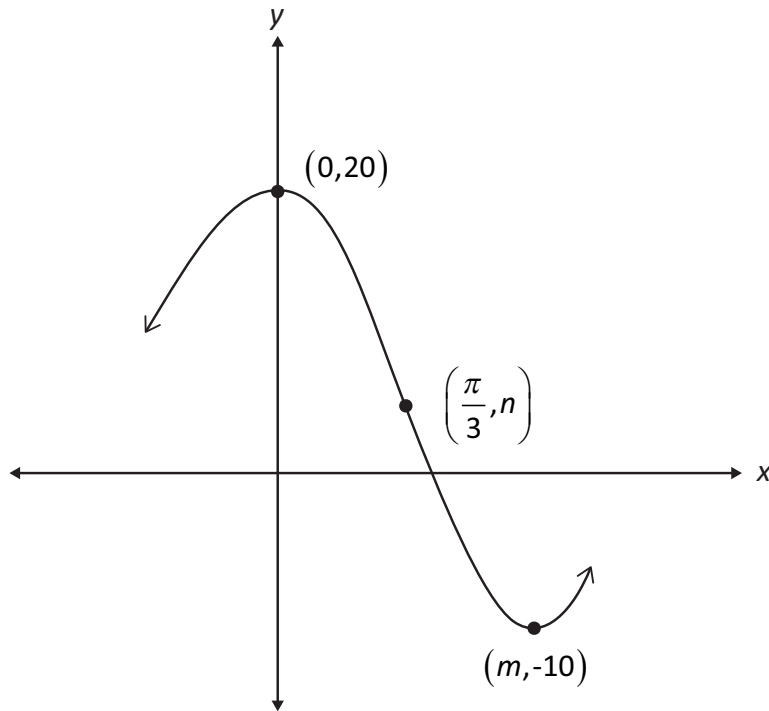
D)



Question 27

T4

The amplitude, A , of the sinusoidal function can be determined using the equation $A = 20 - n$. State the values of m and n .



Solution

$$m = \frac{2\pi}{3}$$

1 mark for value of m

$$n = 5$$

1 mark for value of n

2 marks

Question 28

P2

Solve, algebraically.

$${}_{n-1}P_2 = 20$$

Solution

$$\frac{(n-1)!}{(n-1-2)!} = 20$$

½ mark for substitution

$$\frac{(n-1)!}{(n-3)!} = 20$$

$$\frac{(n-1)(n-2)\cancel{(n-3)!}}{\cancel{(n-3)!}} = 20$$

1 mark for factorial expansion

½ mark for simplification of factorials

$$n^2 - 3n + 2 = 20$$

$$n^2 - 3n - 18 = 0$$

$$(n-6)(n+3) = 0$$

$$n = 6 \quad \cancel{n = -3}$$

½ mark for the permissible value of n

½ mark for showing the rejection of the extraneous root

3 marks

Question 29

T6

Determine the exact value of $\sin\left(\frac{23\pi}{12}\right)$.

Solution

$$\sin\left(\frac{23\pi}{12}\right) = \sin\left(\frac{21\pi}{12} + \frac{2\pi}{12}\right)$$

$$\sin\left(\frac{7\pi}{4} + \frac{\pi}{6}\right) = \sin\left(\frac{7\pi}{4}\right)\cos\left(\frac{\pi}{6}\right) + \cos\left(\frac{7\pi}{4}\right)\sin\left(\frac{\pi}{6}\right)$$

$$= \left(-\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right)$$

$$= \frac{-\sqrt{6} + \sqrt{2}}{4}$$

1 mark for substitution into correct identity

2 marks for exact values (½ mark for each)

3 marks

Note:

Other combinations are possible.

Question 30

R10

Cameron was asked to solve the equation, $\log_3(x-4)=2$.

Cameron's solution:

$$x - 4 = 2^3$$

$$x - 4 = 8$$

$$x = 12$$

Describe Cameron's error.

Solution

Cameron mixed up the exponent and the base when changing to exponential form.

1 mark

Question 31

T6

Given the identity $\sin^2 \theta = \frac{\tan \theta \sin \theta}{\sec \theta}$, state the non-permissible value of θ over the interval $[\pi, 2\pi]$.

Solution

$$\cos \theta = 0$$

$$\theta = \frac{3\pi}{2}$$

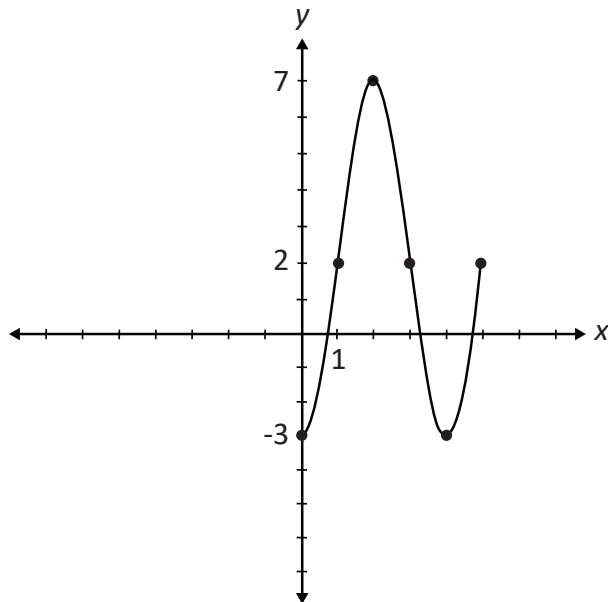
1 mark

Question 32

T4

Sketch the graph of $y = -5\cos\left(\frac{\pi}{2}x\right) + 2$ over the domain $[0, 5]$.

Solution



1 mark for shape of a sinusoidal function with correct amplitude

1 mark for vertical reflection

1 mark for vertical translation

1 mark for period

4 marks

Question 33

R12

Describe the behaviour of the graph of the polynomial function, $p(x) = (x-3)^2(x-1)$, as it approaches the x-intercept at $x=3$.

Solution

As it approaches $x=3$, the graph touches but does not cross the x-axis.

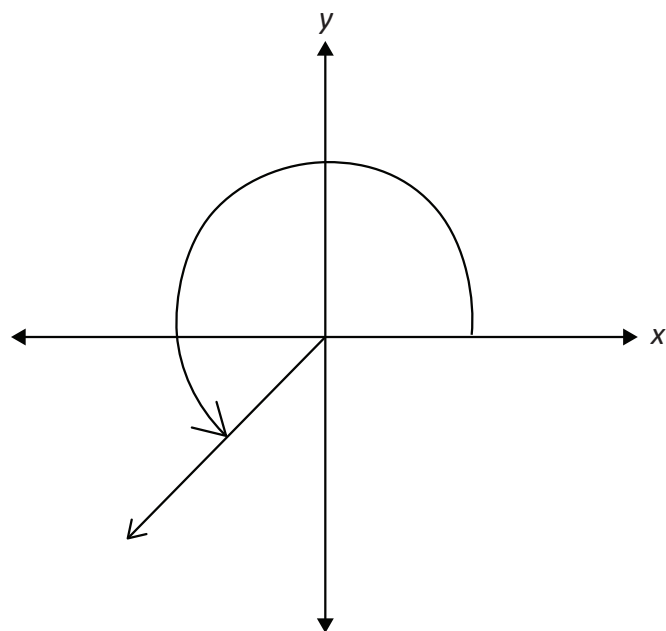
1 mark

Question 34

T1

Sketch the angle of 4 radians in standard position.

Solution



$\frac{1}{2}$ mark for an appropriate angle
in quadrant III

$\frac{1}{2}$ mark for the correct direction of an
angle in standard position

1 mark

Question 35

R10

Solve, algebraically.

$$\log_5 12 + 2\log_5 x = \log_5 48$$

Solution

Method 1

$$\log_5 12 + 2\log_5 x = \log_5 48$$

$$\log_5 12 + \log_5 x^2 = \log_5 48$$

1 mark for power law

$$\log_5 12x^2 = \log_5 48$$

1 mark for product law

$$12x^2 = 48$$

1 mark for equating arguments

$$x^2 = 4$$

$$x = 2 \quad \cancel{x = -2}$$

½ mark for permissible value of x

½ mark for showing the rejection of the extraneous root

4 marks

Method 2

$$\log_5 12 + 2\log_5 x = \log_5 48$$

$$\log_5 12 + \log_5 x^2 - \log_5 48 = 0$$

1 mark for power law

$$\log_5 \left(\frac{12x^2}{48} \right) = 0$$

½ mark for product law

½ mark for quotient law

$$\frac{12x^2}{48} = 5^0$$

1 mark for exponential form

$$x^2 = 4$$

$$x = 2 \quad \cancel{x = -2}$$

½ mark for permissible value of x

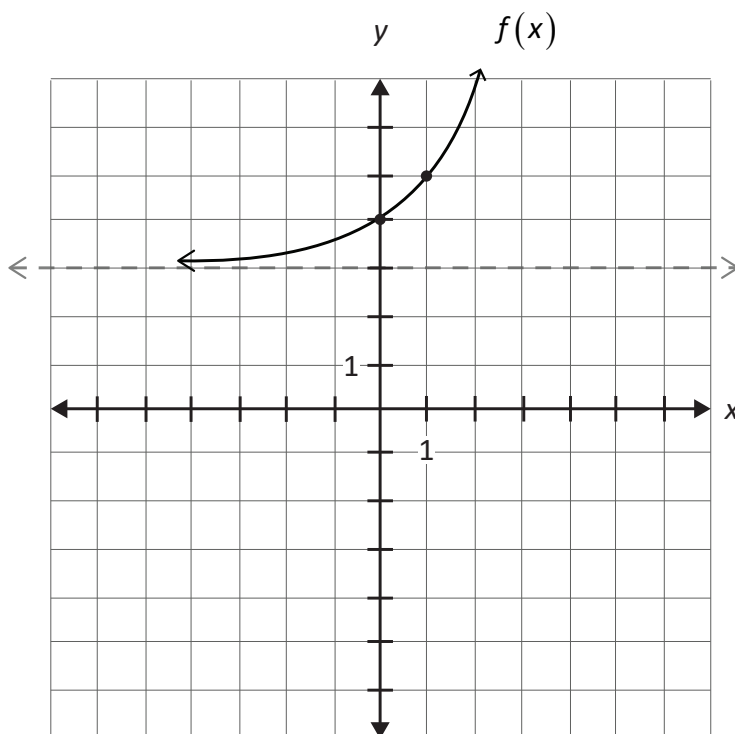
½ mark for showing the rejection of the extraneous root

4 marks

Question 36

R9

State the equation of the exponential function represented by the graph of $f(x)$.



Solution

$$f(x) = 2^x + 3$$

1 mark for exponential function with a base of 2
1 mark for vertical translation

2 marks

Question 37

T3

Evaluate.

$$\cos\left(\frac{11\pi}{3}\right) + \csc\left(-\frac{\pi}{3}\right) \cot\left(\frac{11\pi}{6}\right)$$

Solution

$$\left(\frac{1}{2}\right) + \left(-\frac{2}{\sqrt{3}}\right) \left(-\frac{\sqrt{3}}{1}\right)$$

$$\frac{1}{2} + 2$$

$$\frac{5}{2}$$

1 mark for $\cos\left(\frac{11\pi}{3}\right)$ ($\frac{1}{2}$ mark for value; $\frac{1}{2}$ mark for quadrant)

1 mark for $\csc\left(-\frac{\pi}{3}\right)$ ($\frac{1}{2}$ mark for value; $\frac{1}{2}$ mark for quadrant)

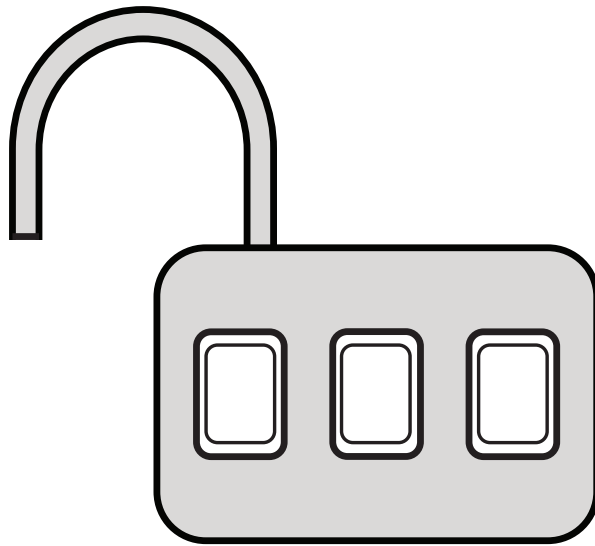
1 mark for $\cot\left(\frac{11\pi}{6}\right)$ ($\frac{1}{2}$ mark for value; $\frac{1}{2}$ mark for quadrant)

3 marks

Question 38

P1

Min Li has a 3-digit code for his lock. Determine the number of possible codes for his lock, if repetition is allowed.



Solution

$$\underline{10} \cdot \underline{10} \cdot \underline{10} = 1000$$

1 mark

Question 39

R10

Solve, algebraically.

$$\left(\frac{1}{6}\right)^{3x+2} = 6^{2x}$$

Solution

$$(6^{-1})^{3x+2} = 6^{2x}$$

1 mark for changing to a common base

$$6^{-3x-2} = 6^{2x}$$

$$-3x - 2 = 2x$$

1 mark for equating exponents

$$-5x = 2$$

$$x = -\frac{2}{5}$$

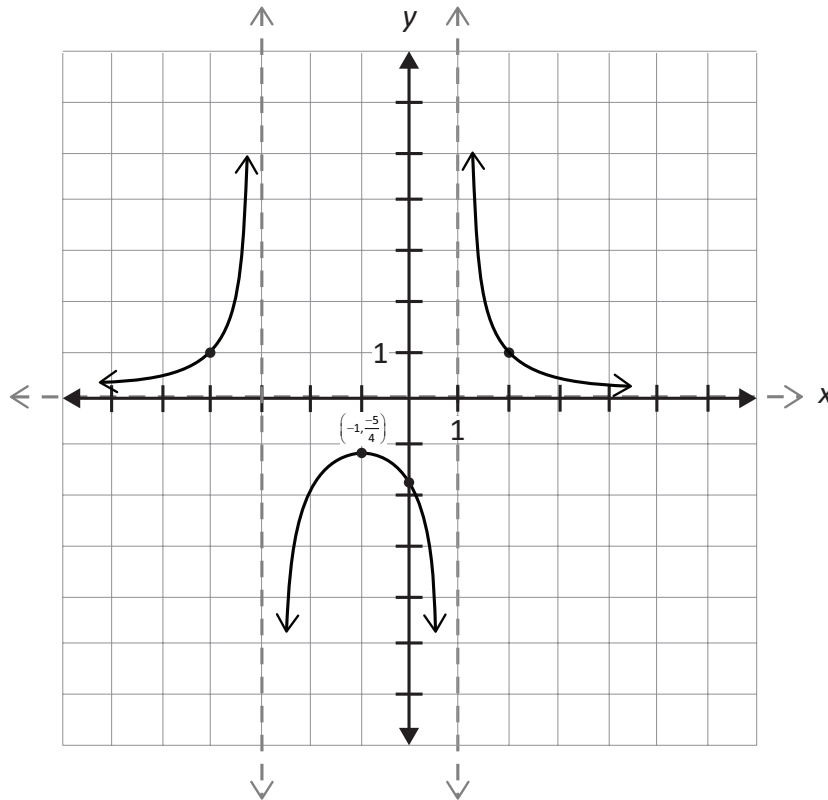
2 marks

Question 40

R14

Sketch the graph of $f(x) = \frac{5}{(x-1)(x+3)}$ and state the y-intercept.

Solution



y-intercept: $\left(0, -\frac{5}{3}\right)$

1 mark for vertical asymptotic behaviour

(½ mark for behaviour approaching $x = -3$; ½ mark for behaviour approaching $x = 1$)

1 mark for horizontal asymptotic behaviour approaching $y = 0$

1½ marks for shape (½ mark for shape of each branch)

½ mark for y-intercept

4 marks

Question 41

R7, R10

Given $\log_2 x = 5$, determine the value of $\log_4 (2x)$.

Solution

$$x = 2^5$$

1 mark for solving for x

$$x = 32$$

$$\log_4 (2(32))$$

$$\log_4 (64)$$

$$3$$

1 mark for evaluating the logarithm

2 marks

Question 42

T6

Henry was asked to prove the identity, $\csc^2 \theta - 2\cos^2 \theta + 1 = \cot^2 \theta + 2\sin^2 \theta$, for all permissible values of θ .

His work:

Left-Hand side	Right-Hand side
$\csc^2\left(\frac{\pi}{4}\right) - 2\cos^2\left(\frac{\pi}{4}\right) + 1$	$\cot^2\left(\frac{\pi}{4}\right) + 2\sin^2\left(\frac{\pi}{4}\right)$
$(\sqrt{2})^2 - 2\left(\frac{\sqrt{2}}{2}\right)^2 + 1$	$(1)^2 + 2\left(\frac{\sqrt{2}}{2}\right)^2$
2 - 1 + 1	1 + 1
2	2
LHS	RHS ✓

Explain why his proof is insufficient.

Solution

Henry proved the identity for $\theta = \frac{\pi}{4}$, but not for all permissible values of θ .

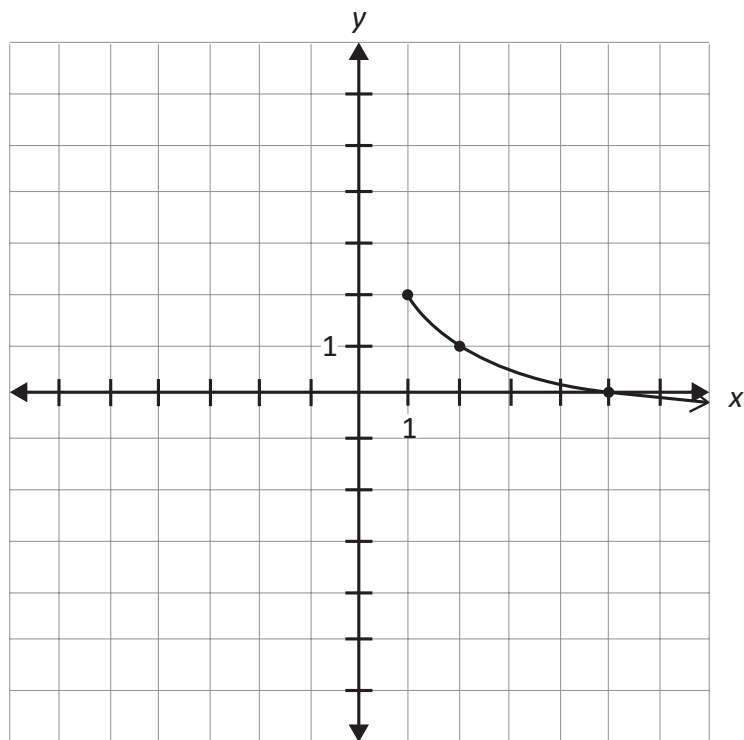
1 mark

Question 43

R13

Sketch the graph of $y - 2 = -\sqrt{x - 1}$.

Solution



1 mark for shape of a radical function
1 mark for vertical reflection
1 mark for horizontal translation
1 mark for vertical translation

4 marks

Question 44

R7, R8

Evaluate.

$$\log_3 54 - \log_3 6$$

Solution

$$\log_3 \left(\frac{54}{6} \right)$$

1 mark for quotient law

$$\log_3 9$$

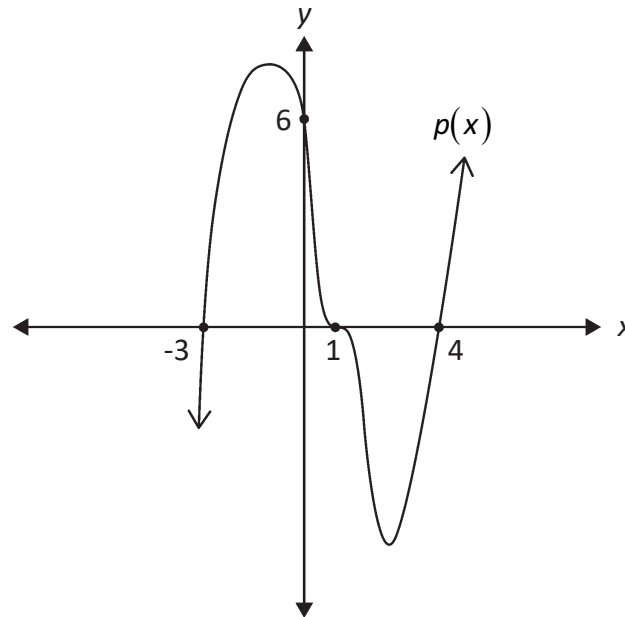
$$2$$

1 mark for evaluating the logarithm

2 marks

Question 45**R12**

Determine, algebraically, the equation of the graph of the polynomial function, $p(x)$.

**Solution**

$$p(x) = a(x-1)^3(x+3)(x-4)$$

½ mark for factors of $p(x)$

½ mark for multiplicity of 3 at $x = 1$

$$p(0) = a(0-1)^3(0+3)(0-4)$$

$$6 = a(-1)^3(3)(-4)$$

$$6 = 12a$$

$$\frac{1}{2} = a$$

1 mark for consistent value of a

$$p(x) = \frac{1}{2}(x-1)^3(x+3)(x-4)$$

2 marks